

Practical exercises

— surface area and volume of prisms

A Cambridge insert: base area times height for the volume, and the net for the surface.

LESSON 168

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 15

LESSON CLOCK

8'

12'

12'

6'

2'

What a prism is	8 min
Volume	12 min
Surface area from the net	12 min
Units and capacity	6 min
Homework	2 min

LEARNING OBJECTIVES

- Use $V = Bh$ for any prism.
- Find the surface area from the net, as $2B + Ph$.
- Work with the units of volume and capacity.
- Solve practical problems about boxes, tanks and channels.

TEXTBOOK

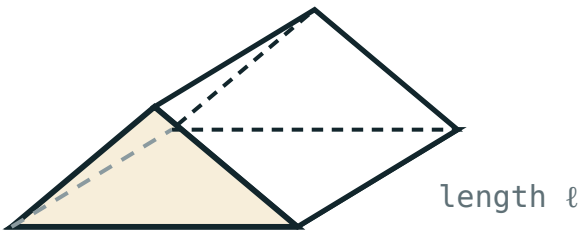
National	pp. 511–514
Cambridge	Stage 8, pp. 148–157
Workbook	Exercise 15.3

01

Explanation

What a prism is

A **prism** has two identical parallel faces — the bases — joined by rectangles. Its cross-section is the same all the way along.



base area S

$$V = S \cdot \ell$$

A prism: the base swept along its height

SOLID	A PRISM?	BASE
cuboid	yes	a rectangle
cube	yes	a square
triangular prism	yes	a triangle
cylinder	yes, in effect	a circle
pyramid	no	it narrows to a point

ONE FORMULA FOR ALL OF THEM

Because the cross-section never changes, the volume is the base area repeated all the way up. That is $V = Bh$, whatever shape the base has.

Volume

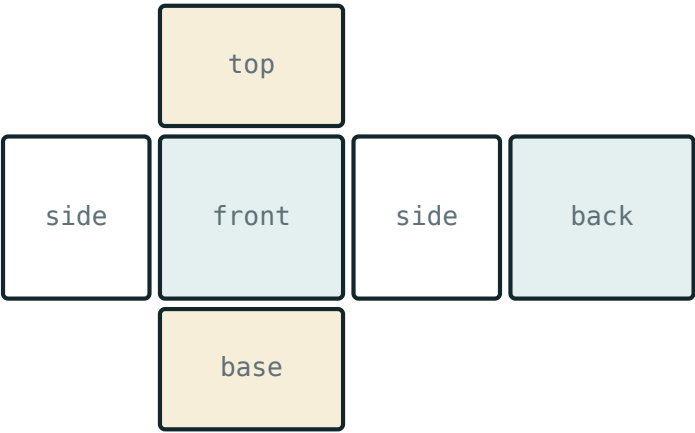
$V = Bh$

PRISM	BASE AREA B	HEIGHT H	VOLUME
cuboid $5 \times 4 \times 3$	20	3	60 cm^3
cube of edge 4	16	4	64 cm^3
triangular, base 6, height 4, length 10	12	10	120 cm^3
cylinder $r = 7, h = 10$	154	10	1540 cm^3

THE HEIGHT IS MEASURED PERPENDICULAR TO THE BASE

For a triangular prism lying on its side, the “height” of the prism is its length, and the triangle's own height belongs to the base area. Two different heights appear, and confusing them is the usual error.

Surface area from the net



surface area = the area of the whole net

The net of a prism: two bases and one long rectangle

$$S = 2B + Ph$$

where P is the perimeter of the base: the two bases, plus one rectangle whose width is the base's perimeter and whose height is the prism's.

PRISM	$2B$	PH	S
cuboid $5 \times 4 \times 3$	40	$18 \cdot 3 = 54$	94 cm^2
cube of edge 4	32	$16 \cdot 4 = 64$	96 cm^2
triangular $3, 4, 5$ base, length 10	12	$12 \cdot 10 = 120$	132 cm^2

A CUBE'S SURFACE IS $6A^2$

Six identical faces — and $2B + Ph$ agrees: $2a^2 + 4a \cdot a = 6a^2$. A formula that survives a special case is a formula worth trusting.

Units and capacity

CONVERSION	FACTOR	WHY
$1 \text{ cm}^3 = 1000 \text{ mm}^3$	10^3	three directions
$1 \text{ litre} = 1000 \text{ cm}^3$	10^3	a 10 cm cube
$1 \text{ m}^3 = 1000 \text{ litres}$	$100^3 \div 1000$	$1\,000\,000 \text{ cm}^3$

PROBLEM	WORKING	ANSWER
a tank $50 \times 40 \times 30$ cm: its capacity	$60\,000\text{ cm}^3$	60 litres
a box $20 \times 20 \times 20$: its volume	8000 cm^3	8 litres
a pool $25 \times 10 \times 2$ m	500 m^3	500 000 litres

LENGTHS SCALE BY k , AREAS BY k^2 , VOLUMES BY k^3

Doubling every edge of a box multiplies its surface by 4 and its volume by 8. This is why a large tank holds far more water per square metre of metal than a small one.

02

Worked examples

EXAMPLE 1 Find the volume and surface area of a cuboid $5 \times 4 \times 3$ cm.

1

$B = 5 \cdot 4 = 20\text{ cm}^2, h = 3.$

2

$V = 60\text{ cm}^3$

3

$P = 2(5 + 4) = 18\text{ cm}, \text{ so } Ph = 54.$

4

$S = 2 \cdot 20 + 54 = 94\text{ cm}^2$

ANSWER

60 cm^3 and 94 cm^2

EXAMPLE 2 A triangular prism has a base triangle with sides 3, 4, 5 cm and length 10 cm. Find its volume and surface area.

① The triangle is right-angled: $B = \frac{1}{2} \cdot 3 \cdot 4 = 6 \text{ cm}^2$.

② $V = 6 \cdot 10 = 60 \text{ cm}^3$

③ $P = 3 + 4 + 5 = 12$, so $Ph = 120$.

④ $S = 12 + 120 = 132 \text{ cm}^2$

ANSWER

60 cm^3 and 132 cm^2

EXAMPLE 3 A tank measures $50 \times 40 \times 30 \text{ cm}$. How many litres does it hold?

① $V = 50 \cdot 40 \cdot 30$

② $= 60\,000 \text{ cm}^3$

③ $60\,000 \div 1000$

④ $= 60 \text{ litres}$.

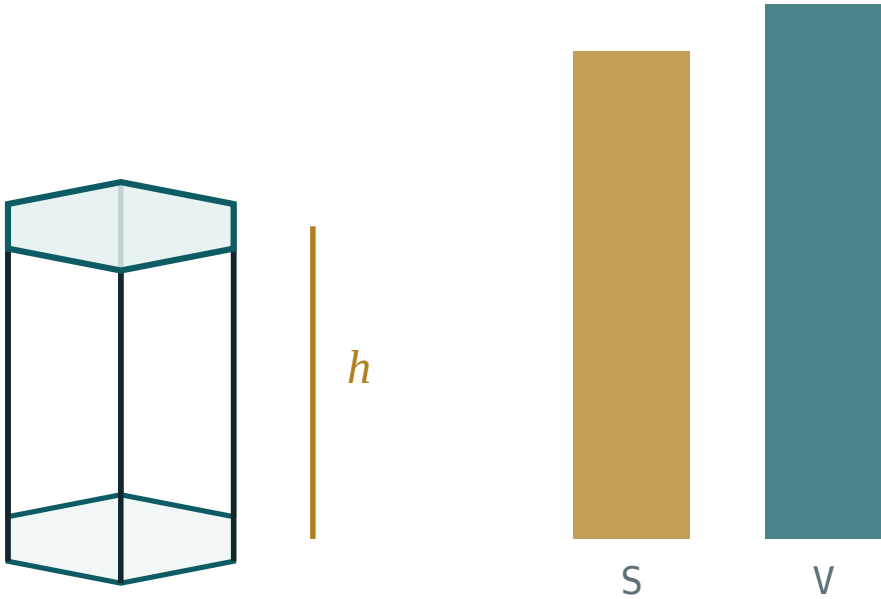
ANSWER

60 litres

03 Interactive model

Cut open a cardboard box along its edges and lay it flat; the surface-area formula is then read straight off the net rather than memorised.

Base area and height decide the volume



base area B **41.6**

perimeter P **24**

total S **227.1**

volume V **249.4**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Prism	Prizma	Призма
Base area	Asos yuzasi	Площадь основания
Height	Balandlik	Высота
Volume	Hajm	Объём
Surface area	To'la sirt	Площадь поверхности
Net	Yoyilma	Развёртка
Capacity	Sig'im	Вместимость
Litre	Litr	Литр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of any prism is:

Bh

$\frac{1}{3} Bh$

$2B + Ph$

Ph

Its surface area is:

Bh

$2B + Ph$

$6a^2$

$2Ph$

A cuboid $5 \times 4 \times 3$ has volume:

12

20

60

94

And surface area:

60

94

96

120

1 litre equals:

100 cm³

1000 cm³

10 000 cm³

1 000 000 cm³

Doubling every edge multiplies the volume by:

2

4

6

8

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A cuboid $5 \times 4 \times 3$: its volume

ANSWER 60 cm^3

2. Its surface area

ANSWER 94 cm^2

3. A cube of edge 4: its volume

ANSWER 64 cm^3

4. Its surface area

ANSWER 96 cm^2

5. A cube of edge a : its surface area

ANSWER $6a^2$

6. 1 litre in cm^3

ANSWER 1000

7. 1 m^3 in litres

ANSWER 1000

Medium · the standard question

7 PROBLEMS

1. A triangular prism, base 6, height 4, length 10

ANSWER 120 cm^3

2. A 3, 4, 5 prism of length 10: its surface area

ANSWER 132 cm^2

3. A cylinder $r = 7$, $h = 10$ ($\pi = \frac{22}{7}$)

ANSWER 1540 cm^3

4. A tank $50 \times 40 \times 30 \text{ cm}$: its capacity

ANSWER 60 litres

5. A box $20 \times 20 \times 20$: its capacity

ANSWER 8 litres

6. A pool $25 \times 10 \times 2 \text{ m}$

ANSWER 500 m^3

7. That pool in litres

ANSWER 500 000

Hard · stretch and reason

7 PROBLEMS

1. Doubling every edge multiplies the surface by

ANSWER 4

2. And the volume by

ANSWER 8

3. A cuboid of volume 120 and base 5×4 : its height

ANSWER 6

4. A cube of volume 125 cm^3 : its surface area

ANSWER 150 cm^2

5. A cube of surface 54 cm^2 : its volume

ANSWER 27 cm^3

6. A channel with trapezium cross-section $B = 15 \text{ cm}^2$ and length 4 m

ANSWER 6000 cm^3

7. Why does a big tank need less metal per litre?

ANSWER Surface grows as k^2 but volume as k^3

07 Homework

Homework — 5 tasks

cm^3 for a volume, cm^2 for a surface — write the units every time.

- Find the volume and surface area of a cuboid $8 \times 5 \times 4 \text{ cm}$.
- Find the volume and surface area of a cube of edge 6 cm.
- A triangular prism has base 8 cm, triangle height 5 cm and length 12 cm. Find its volume.
- A tank measures $80 \times 50 \times 40 \text{ cm}$. How many litres does it hold?
- A cube has volume 216 cm^3 . Find its edge and its surface area.